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Synchrotron X-ray diffraction analysis of 2D residual stress distribution in the brazed joint of PCBN ceramic and M42 steel

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Outline

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Highlights

- Synchrotron XRD first revealed 2D residual stress distribution in PCBN/M42 joints.
- Unexpected tensile residual stress localized at the ceramic-brazed interface.
- M42 steel phase transformation identified as a key factor of this tensile stress.

Abstract

Residual stress is a critical issue in ceramic/metal composite structure joints, yet reports on residual stresses in ceramic/phase-transforming metal joints remain scarce. In this study, the 2D residual stress distribution in a PCBN ceramic and M42 high-speed steel brazed joint is revealed for the first time using synchrotron XRD. The joint comprises a PCBN ceramic, a CuNi interlayer, and M42 steel. Notably, an abnormal residual stress distribution, with tensile stress present at both the surface and interface of the ceramic, is observed for the first time. The residual tensile stress near the ceramic interface exceeds 1000MPa, and joint fracture occurs in the region of maximum tensile stress at the ceramic interface. Finite element analysis indicates that shrinkage of the CuNi layer dominates in joints formed below the phase transition temperature, resulting solely in residual compressive stress at the ceramic interface. In contrast, high-temperature XRD analysis and thermal expansion coefficient measurements confirm that the volume expansion caused by the phase transformation of M42 steel is a key factor contributing to the tensile stress at the ceramic interface.

CRedit authorship contribution statement

Declaration of competing interest

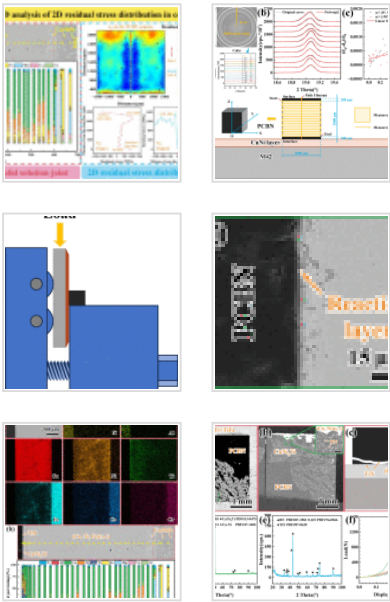
Acknowledgments

Appendix A. Supplementary data

Data availability

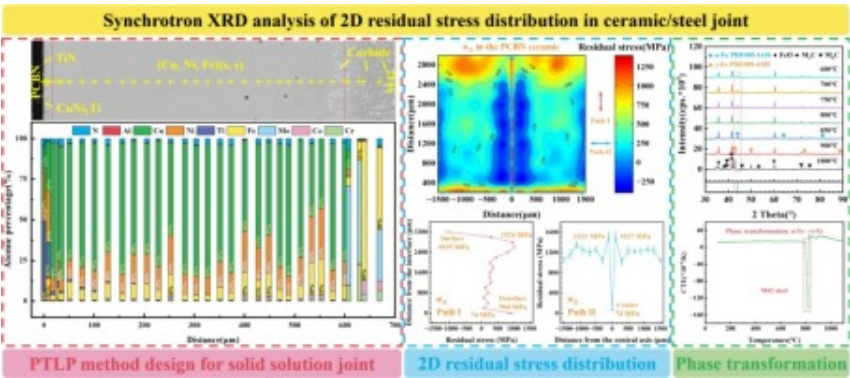
References

Figures (12)



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Graphical abstract



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Keywords

Tables (1)

Table 1. Lattice constants and elastic co...

Braze; Residual stress; Synchrotron XRD; PCBN ceramic; Phase transformation

Extras (1)

 Supplementary material[< Previous article in this issue](#)[Next article in this issue >](#)

1. Introduction

Residual stress is crucial in assessing the reliability of dissimilar material joints, as it can weaken the joint and sometimes lead to cracking due to stress concentration. Thus, research on residual stress distribution of dissimilar joint has drawn great interest [[1], [2], [3]]. Most studies use finite element models (FEM) to analyze residual stress distribution, predicting its magnitude based on the physical properties of the base material and brazing filler [4,5]. These studies often indicate that maximum residual stress occurs at the ceramic interface, making it the most fracture-prone region. However, FEM models frequently overlook complex reaction products and phase transformations in materials, making accurate modeling challenging [6,7]. Therefore, direct experimental measurement of residual stress is essential. The contour method, Raman spectroscopy, and X-ray diffraction are commonly used non-destructive techniques for measuring residual stresses in materials [[8], [9], [10], [11]]. For instance, Muhammad et al. [12] utilized the contour method to measure surface residual stresses in Al/Mg friction stir welded joints, with results aligning well with XRD measurements. Similarly, Geng et al. [13] applied this technique to reconstruct internal residual stresses in Al/steel friction stir welds. Nevertheless, machining ceramic-metal joints poses challenges that complicate contour method implementation, while Raman spectroscopy lacks efficacy for depth-resolved stress analysis in opaque materials. In contrast, X-ray diffraction can penetrate materials like PCBN and graphite with lower attenuation coefficients [14,15]. While some studies have measured residual stress distribution as a function of depth, no reports have yet examined the two-dimensional (2D) residual stress distribution in dissimilar material

joints. Understanding residual stress distribution is vital for optimizing brazing processes and improving joint reliability.

Polycrystalline cubic boron nitride (PCBN) ceramics exhibit exceptional hardness and wear resistance [16], but their brittleness and machining challenges limit broader applications [17]. AISI M42 (W2Mo9Cr4VCo8) high-speed steel provides excellent red hardness and toughness [18,19]. Joining PCBN ceramics with M42 steel in ceramic-metal composite cutting tools balances toughness and hardness, enabling precise and complex machining.

In this study, we employed a partial transient liquid phase (PTLP) method to join PCBN ceramics with M42 high-speed steel. The PTLP bonding process operates through two sequential mechanisms: (1) melting of low-melting-point components to form a transient liquid phase, followed by (2) atomic diffusion-driven isothermal solidification. This two-step mechanism effectively mitigates the formation of continuous brittle intermetallic compounds at the interface [20]. We analyzed the distribution of elements and phases in the brazed joint and measured the 2D residual stress distribution in the PCBN/M42 brazed joint using synchrotron X-ray diffraction (XRD). We also explored the relationship between residual stress distribution, the CTE, and the phase transformation of the base materials, clarifying the connection between the maximum residual stress at the ceramic interface and the fracture location of the joint.

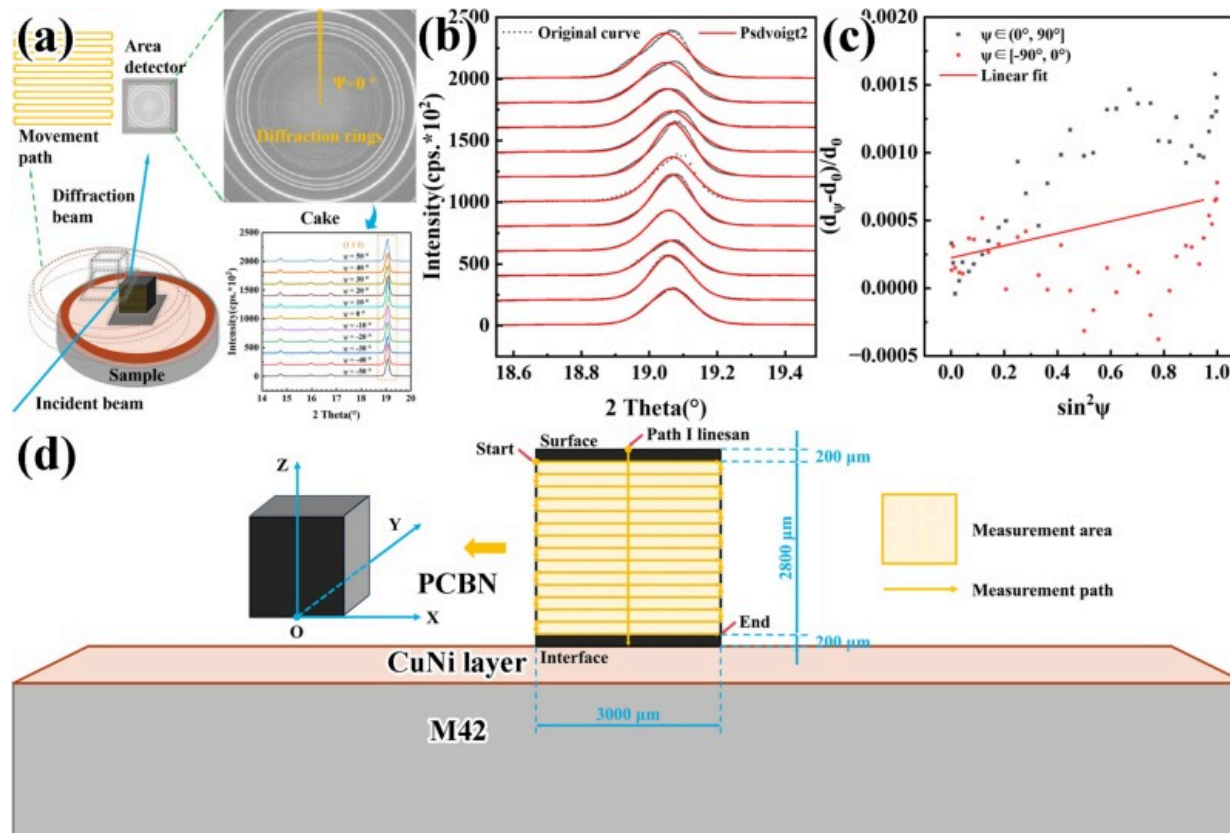
2. Material and experimental methods

The materials used in this study include PCBN ceramics ($3\times3\times3\text{ mm}^3$, Hebei Luohong Technology Co., Ltd. Hebei, China), M42 steel ($\Phi20\times2\text{ mm}^3$, Shanghai Ruiru Metal Group, Co., Ltd. Shanghai, China), Ti foil (10 μm , purity 99.9%, commercial), Cu powder, and Ni powder ($\sim50\mu\text{m}$, purity 99.9%, Beijing Xingrongyuan Technology Co., Ltd. Beijing, China.). The ceramic and steel surfaces were polished to 1 μm using a diamond grinding disc and cleaned in ethanol via ultrasonic cleaning for 15 min. Cu and Ni powders were mixed in an 86:14 mass ratio and pressed into small blocks ($\Phi20\times1\text{ mm}^3$) using a hydraulic press at

300MPa. As shown in Fig. S1, the Cu—Ni block was placed on the M42 steel surface and heated in a vacuum furnace at 1180°C for 10min to form a barrier layer and the details have been reported elsewhere [21]. After furnace cooling, the barrier layer was polished to 1μm finish with a thickness of 600μm, and then cleaned. PCBN ceramic, Ti foil, and M42 steel with the barrier layer were stacked. A 0.1 MPa pressure was applied to ensure tight contact. The workpiece was then heated to 1080°C in a vacuum furnace for 30min. For the CuNi pre-melted layer, the liquidus line of the mixed powder is 1170°C, ensuring full melting of the powder to form a continuous barrier layer under the 1180°C processing parameter. In the PTLP bonding process, the metal adjacent to the interface undergoes three sequential stages: eutectic liquid phase formation, elemental diffusion, and isothermal solidification. According to the Cu-Ni-Ti ternary phase diagram, the temperature and holding time during isothermal solidification govern the formation of intermetallic compounds at the interface. At 1080°C, the interface primarily consists of a $\gamma(\text{Cu})$ solid solution with minor Ti—Ni and Ti—Cu compounds. A holding time of 30min at this temperature ensures sufficient elemental diffusion to achieve complete isothermal solidification.

2D residual stress distribution in PCBN ceramics was measured using synchrotron XRD in transmission geometry at BL14B, Shanghai synchrotron radiation facility (SSRF). An area detector was used to collect diffraction signals, allowing diffraction rings corresponding to different crystal planes to be obtained. As presented in Fig. 1(a), the incident angle was fixed at zero, and the spot size was adjusted to 200μm×200μm. Initially, the light beam was directed at the left edge of the PCBN ceramic surface. The sample was then moved in a serpentine pattern with a step size of 200μm, causing the interaction volume between the light beam and the PCBN to gradually shift toward the interface. The peak measurement of the (111) crystal plane was selected, residual stress was calculated using the $\sin^2 \psi$ method [22], with peak fitting performed using DAWN and Origin software [[23], [24], [25]]. To enhance accuracy, the integration area is set to 2°. Fig. 1(d) presents an enlarged front-view schematic of the sample measurement procedure shown in Fig. 1(a). In this view, the yellow-shaded region denotes the total X-ray scanning area

(3000 μm ×2800 μm), while the rectangular box is positioned 200 μm away from both the ceramic surface and the interface. The yellow line indicates the scanning trajectory, with Path I representing an additional line scan along the central axis extending from the ceramic surface to the interface. The sample coordinate system (illustrated in the left panel) defines the Z-axis as perpendicular to the ceramic surface, with the X-ray incident direction aligned parallel to the Y-axis—indicating that the measurement region lies within the X-Z plane.



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Fig. 1. (a) Schematic of the geometry of the residual stress measurement, the synchrotron X-ray fully penetrate through the PCBN ceramic and the diffraction pattern is recorded by

the area detector; (b) integral curve extracted using DAWN software and peak fitting using PseudoVoigt2 model; (c) linear fitting between $(d_\psi - d_0)/d_0$ and $\sin^2 \psi$; (d) enlarged front-view schematic of the sample measurement procedure shown in Fig. 1(a).

Due to the joint's symmetry in both X and Y directions, the residual stress in PCBN was considered biaxial plane stress. For the $\sin^2 \psi$ method, the relationship between the residual stress σ_{11} and $\sin^2 \psi$ is expressed as [26]:

$$\varepsilon_\psi = (d_\psi - d_0)/d_0 = 1/2 \bullet S_2 \sigma_{11} \sin^2 \psi + S_1 (\sigma_{11} + \sigma_{22}) \quad (1)$$

where d_ψ is the d-spacing measured at angle ψ , and d_0 is the stress-free d-spacing. Since d_0 is difficult to determine, the d-spacing at $\psi = 0$ is used instead, without introducing significant error. $1/2 \bullet S_2$ is the X-ray elastic constant, calculated using ISODEC software [27], and set to $1.104291 \times 10^{-6} \text{MPa}^{-1}$. The residual stress is the average stress within the X-ray interaction volume, where the volume size is defined by the spot dimensions ($200\mu\text{m} \times 200\mu\text{m}$) multiplied by the penetration depth (3mm), i.e., a total volume of $200\mu\text{m} \times 200\mu\text{m} \times 3\text{mm}$. In the sample coordinate system, σ_{11} denotes the X-direction normal stress (i.e., σ_x), and σ_{22} corresponds to the Y-direction normal stress (i.e., σ_y). Both components are in-plane stresses. The parameters are listed in Table 1 [28]. Residual stress is determined from the slope between $(d_\psi - d_0)/d_0$ and $1/2 \bullet S_2 \sin^2 \psi$, calculated using the linear fitting tool in Origin software.

Table 1. Lattice constants and elastic constants of PCBN ceramics used in ISODEC software [28].

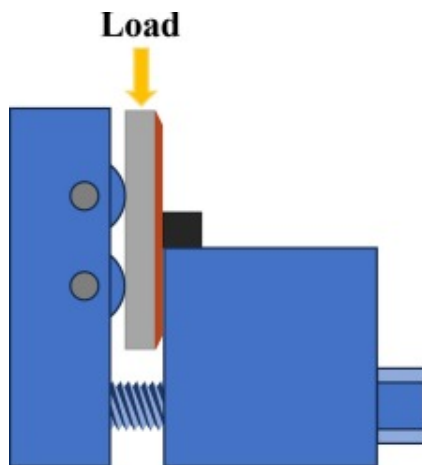
a(Å)	b(Å)	c(Å)	C11(GPa)	C22(GPa)	C44(GPa)
3.617	3.617	3.617	828.2	201.2	492.3

After measuring residual stress, the brazed joint is cut along the longitudinal section with a diamond wire cutting machine and polished to $1\mu\text{m}$ with a diamond grinding disc. The

microstructure and elemental distribution are examined using a scanning electron microscope (SEM, Zeiss, OR, USA) coupled with energy dispersive X-ray spectroscopy (EDS). Shear strength was determined via shear testing, with the test setup schematic illustrated in Fig. 2. Specifically, the thread length was adjusted to ensure the specimen was subjected to a pure shear state during loading, conducted at a constant rate of 0.5 mm/min. The shear strength (σ_s) was calculated using the formula:

$$\sigma_s = F/S \quad (2)$$

where F is the applied load (in N) and S is the shear area (in mm²).



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Fig. 2. Schematic diagram of sample shear test.

The phase composition of the fracture surface is analyzed using XRD (D8-Advance, Karlsruhe, Germany). High-temperature XRD characterizes phase changes in M42 steel and the Cu—Ni barrier layer, while their coefficient of thermal expansion (CTE) values are measured with a thermal expansion instrument.

3. Results and discussion

Fig. 3 shows the microstructure of PCBN ceramic and M42 steel brazed with a 10 μm Ti foil at 1080 $^{\circ}\text{C}$ for 30min. The interfaces are well-bonded, with no visible holes or cracks. The brazed seam, including a CuNi barrier layer, is about 600 μm wide. Z-contrast in the central brazed joint is uniform, with no intermetallic compounds, indicating a solid solution brazing seam has been achieved. A continuous reaction layer, about 1.5 μm thick, forms at the ceramic interface, with small gray granular compounds nearby. A continuous diffusion layer, around 50 μm thick, is observed at the M42 steel interface, with white compounds distributed within it.



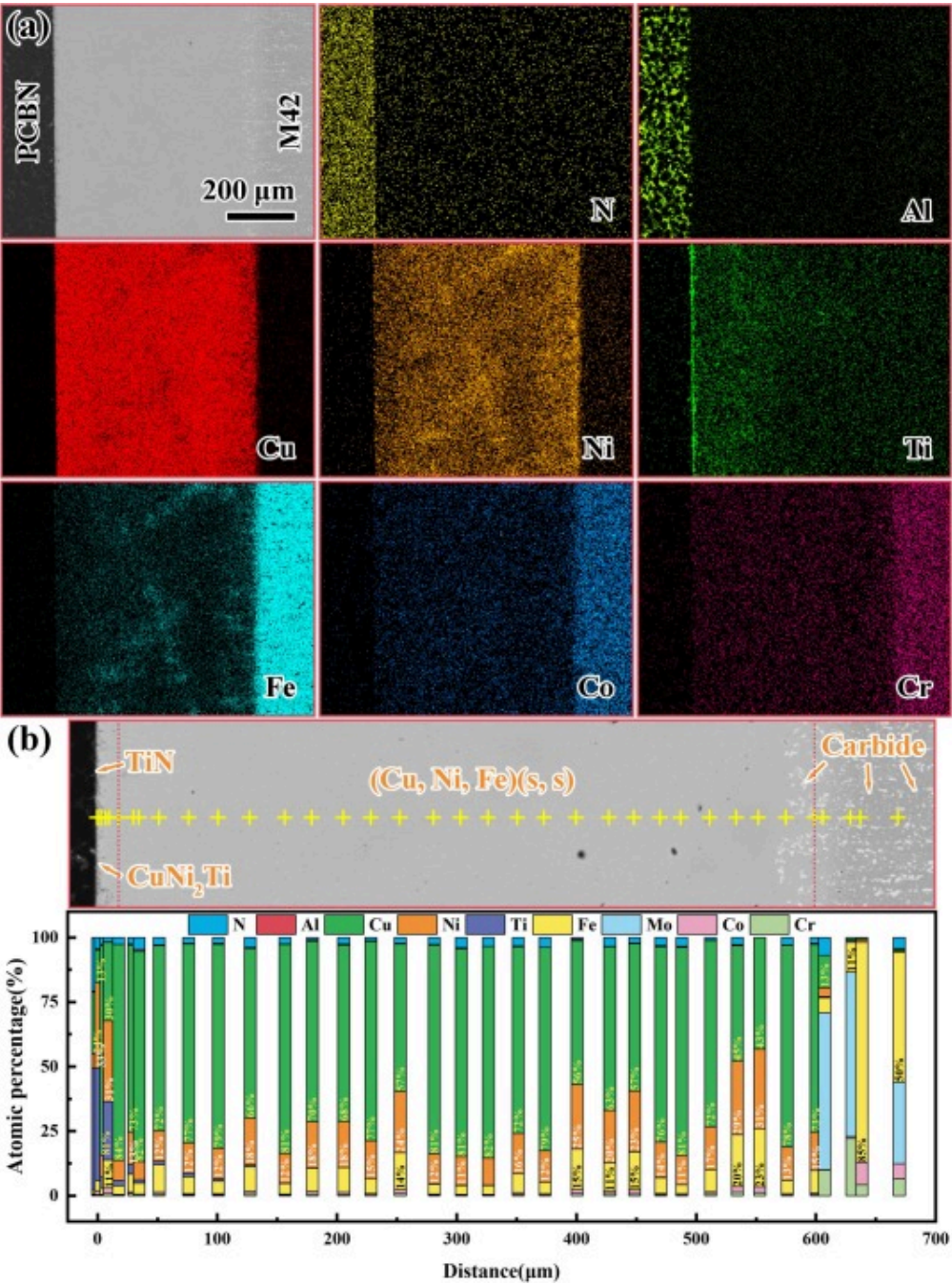
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Fig. 3. (a) Microstructure of the PCBN-M42 brazed joint; (b) enlarged image of the PCBN ceramic side; (c) enlarged image of the M42 steel side.

EDS mapping in Fig. 4 shows that the CuNi layer acts as a barrier to Ti diffusion. Ti is concentrated at the ceramic interface, while Cu and Ni are uniformly distributed in the brazed seam. The diffusion layer on the M42 steel side contains Cu, Ni, and Fe, indicating Fe dissolves into the seam. The atomic percentages in different regions are shown in Fig. 4(b). The continuous reaction layer at the ceramic interface has a high Ti and N content, corresponding to the TiN phase. The gray phase near this layer contains Cu, Ni, and Ti, suggesting CuNi_2Ti and $\text{Ti}(\text{Cu}, \text{Ni})_2$ intermetallic compounds [21,29]. The region about 20 μm thick near the reaction layer is over 80% Cu, indicating a Cu-based solid solution. Further away from the reaction layer, the region extending into the diffusion layer shows higher proportions of Cu, Ni, and Fe, forming a (Cu, Ni, Fe) solid solution. The white phase

in the diffusion layer and M42 steel matrix, rich in Mo and Cr, is speculated to be carbides, with larger particles likely being primary carbide M_2C and smaller ones secondary carbide M_6C [30].

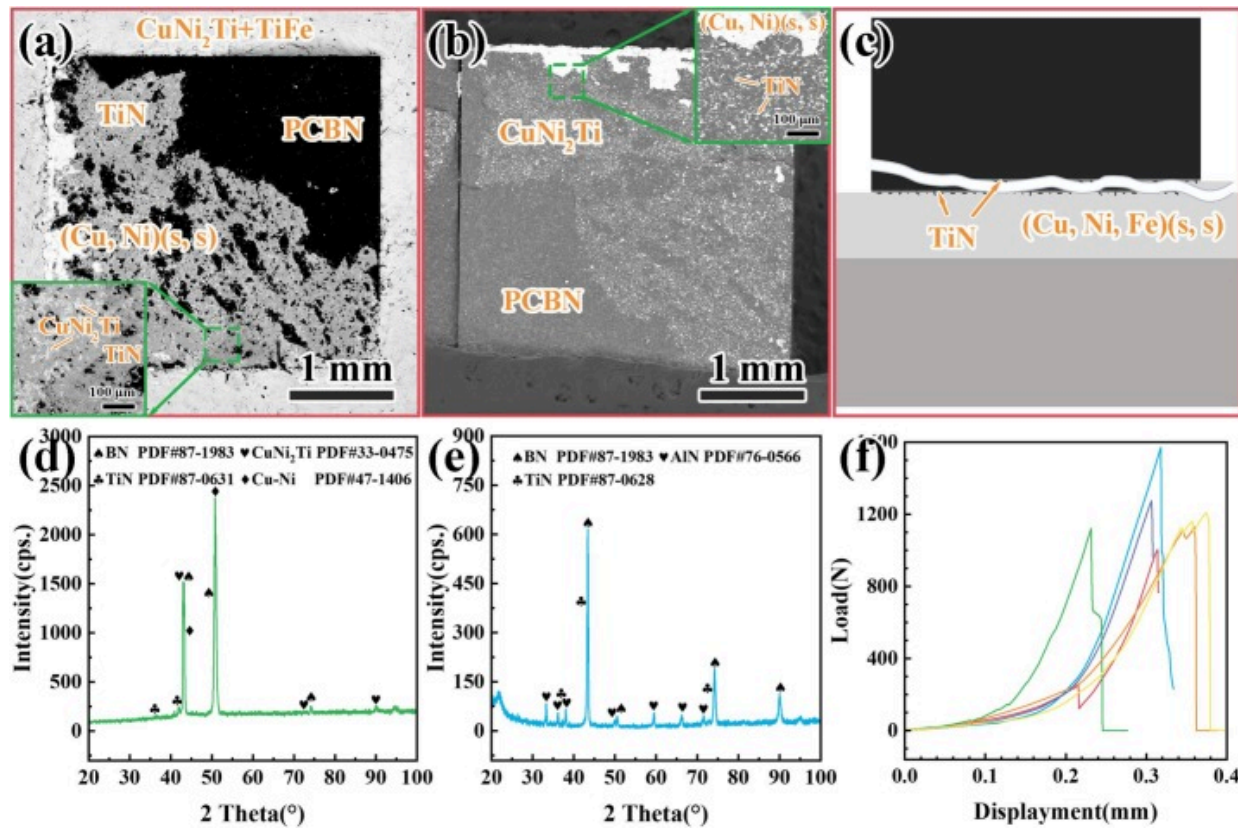


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Fig. 4. (a) Distribution of elements within the PCBN-M42 joint; (b) atomic percentage and primary phases at each point within the joint.

The fracture morphology of the joint after the shear test is shown in [Fig. 5](#). A small amount of ceramic phase is observed, and the cross-section appears flat. The ceramic is fractured approximately 200 μm from the interface. The fracture surface on M42 steel side consists of dark gray phase, light gray phase, and meanwhile white phase at the ceramic edge. EDS analysis identifies these as TiN, CuNi_2Ti , and $(\text{Cu, Ni})(\text{s, s})$, respectively. The bright white phase at the ceramic interface is also CuNi_2Ti . Based on the brazed joint microstructure in [Fig. 4](#), the fracture occurred near the interface between the ceramic and the $(\text{Cu, Ni, Fe})(\text{s, s})$. [Fig. 5\(c\)](#) shows a schematic of the fracture path. XRD results in [Fig. 5\(d\)](#) and (e) confirm the presence of BN, TiN, and CuNi_2Ti at the fracture site. [Fig. 5\(f\)](#) presents the displacement-load curve of the shear test, revealing a characteristic brittle fracture. As the external load increases, cracks initially form at the ceramic edge near the interface and then rapidly propagate along the interface between the ceramic and the brazed seam, ultimately leading to joint failure. This suggests that the ceramic near the interface is the weakest region of the joint and may be subject to significant residual stress. Nonetheless, the average shear strength of the joint is 132 MPa, which is higher than the shear strength of using AgCuTi brazing filler (~ 80 MPa).



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Fig. 5. (a) Fracture morphology and primary phases on the M42 steel side; (b) fracture morphology and primary phases on the PCBN ceramic side; (c) schematic diagram of the fracture path through the joint; (d) XRD pattern of the steel side fracture; (e) XRD pattern of the ceramic side fracture; (f) displacement-load curve of the joint.

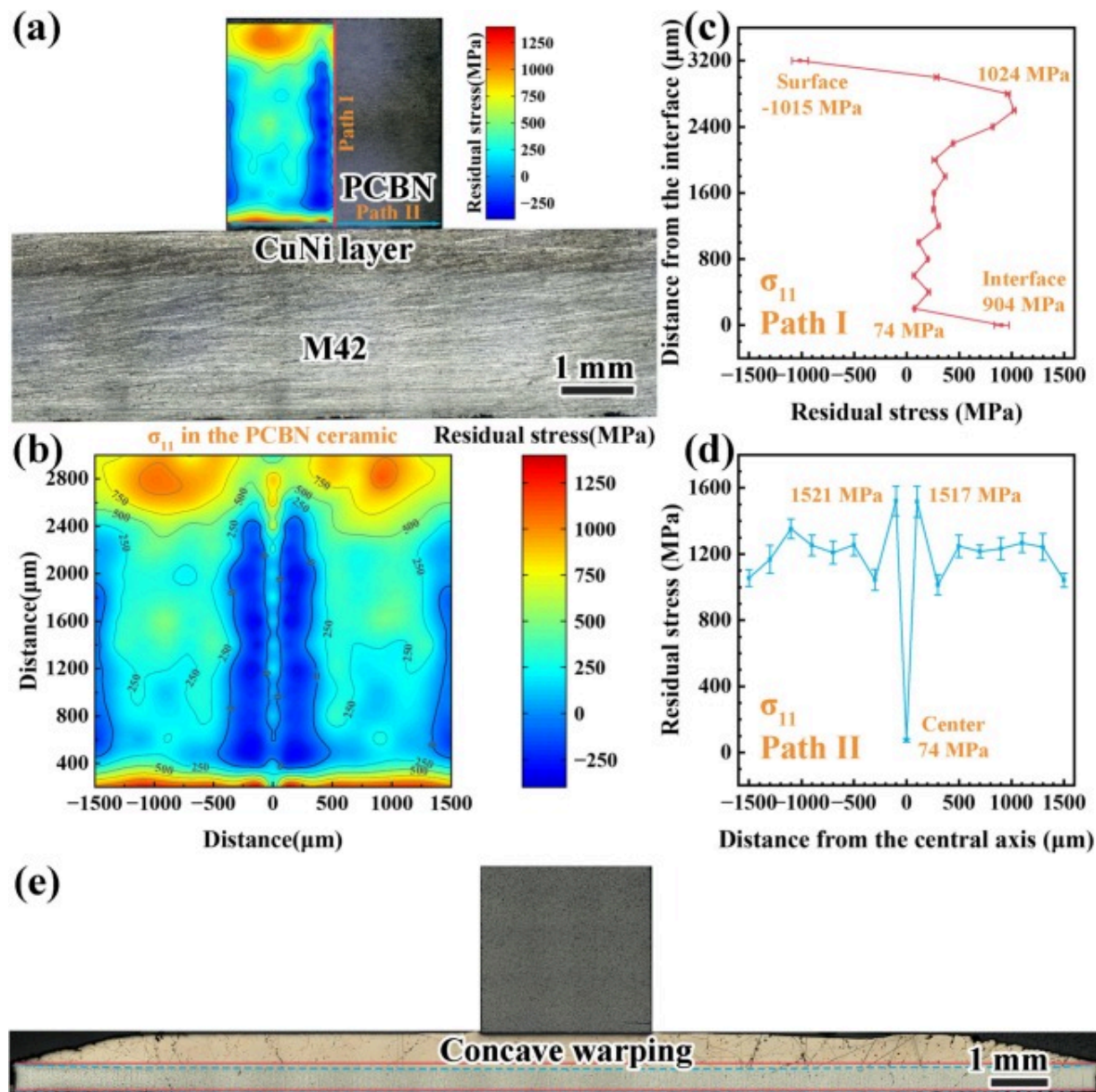
As illustrated in Fig. 1, residual stress was measured using a transmission geometry setup. The absorption of X-rays in the material was calculated based on the Beer-Lambert law [31]:

$$I/I_0 = e^{-\mu/\rho \cdot \rho t} \quad (3)$$

where I_0 represents the initial intensity of the X-ray beam, I is the intensity after passing through the material, and μ/ρ is the mass attenuation coefficient for X-rays at 18keV, which is $0.651 \text{ cm}^2/\text{g}$ for cubic boron nitride [15]. The material's density ρ is 3.49 g/cm^3 , and τ the distance the X-ray beam travels through the material. For a 1% absorption limit, calculations reveal that the X-ray beam can penetrate cubic boron nitride to a thickness exceeding 20mm, ensuring full sample penetration.

The peak was fitted using the PseudoVoigt2 model, achieving an adjusted R-squared value of 0.98. A linear fit of $(d_\psi - d_0)/d_0$ versus $\sin^2 \psi$ was performed, with one fitting result displayed in Fig. 1(c). The positive slope indicates tensile residual stress near the interface. For clarity, the “interface” refers to the ceramic-brazed seam boundary, while the “surface” is the ceramic's upper layer. The 2D residual stress distribution of the joint is presented in Fig. 6. Quite surprisingly, residual stresses near the ceramic interface and surface are high and tensile, while compressive stresses appear near the sides and central axis. The detailed residual stress values are provided in Table S1. An enlarged view in Fig. 6(b) reveals a symmetrical stress distribution, with tensile stress observed within $400 \mu\text{m}$ of the interface and between 200 and $800 \mu\text{m}$ from the surface. Compressive stress is present within 100 – $500 \mu\text{m}$ of the central axis and $200 \mu\text{m}$ from both sides, opposite to the trend in conventional ceramic brazed joints [[32], [33], [34]]. The maximum tensile stress, exceeding 1000 MPa , occurs $200 \mu\text{m}$ from the interface and $400 \mu\text{m}$ from the surface, aligning with the ceramic fracture site. The maximum compressive stress, over 250 MPa , appears at $200 \mu\text{m}$ from the central axis. Residual stress along paths I and II, shown in Fig. 6(c) and (f), further highlight these patterns. Path I (central axis) shows tensile stress, except at the surface, where compressive stress reaches -1015 MPa . The maximum tensile stress of 1024 MPa occurs $600 \mu\text{m}$ from the surface. Path II (parallel to the interface, $200 \mu\text{m}$ away) shows entirely tensile stress, ranging from 74 MPa at the center to $\sim 1500 \text{ MPa}$ $100 \mu\text{m}$ from the central axis. Tensile stress at the sides also exceeds 1000 MPa . These results confirm that residual stress at the brazed joint interface critically affects mechanical properties. Further evidence comes from wire electrical discharge machining (EDM) analysis of the brazed PCBN/M42 joint. After thinning the M42 steel

specimen, the joint exhibits concave deformation due to insufficient stiffness constraints in the thickness direction (as illustrated: the red line represents the baseline straight reference, while the blue line denotes the deformed profile). This confirms the presence of tensile stress at the ceramic interface—had compressive stress been present, the interface would have displayed convex deformation instead.



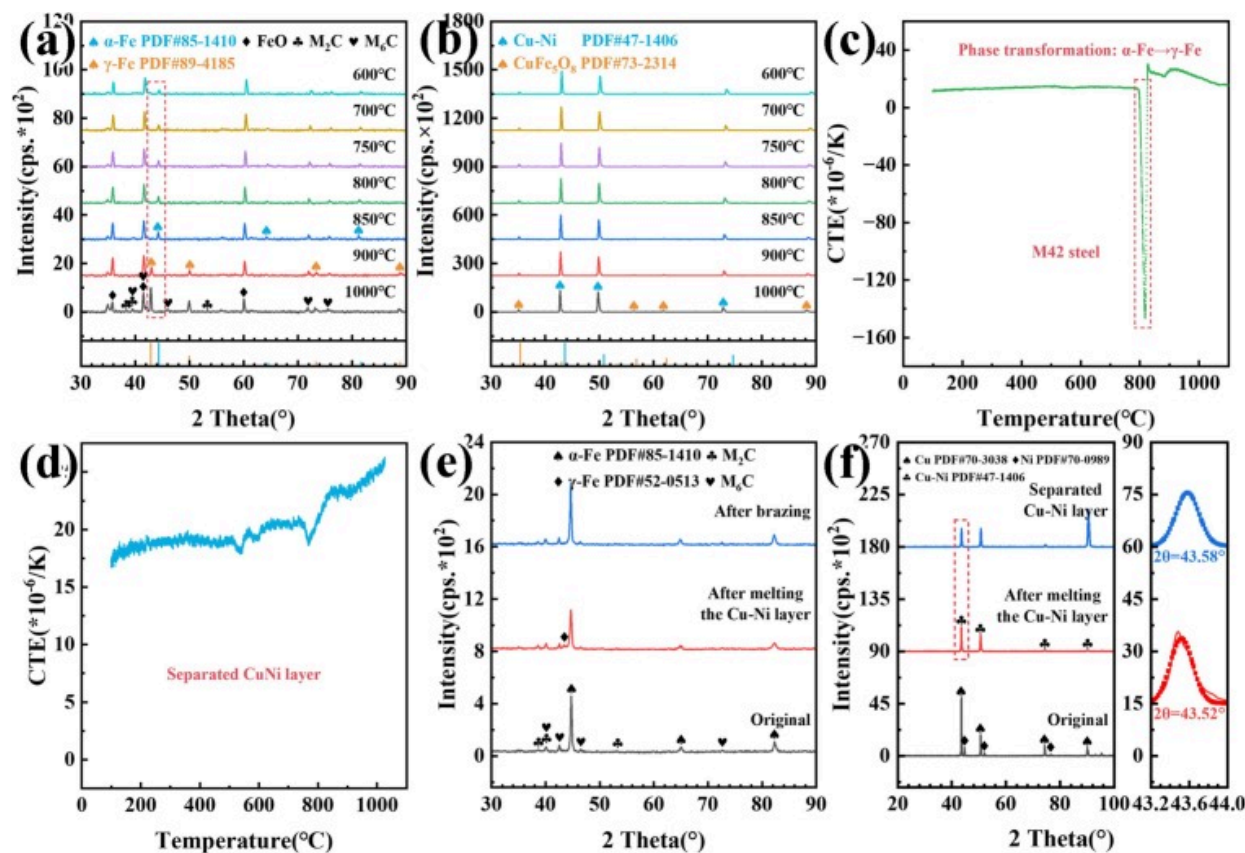
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Fig. 6. (a) Image indicating the location of residual stress measurement in the brazed joints; (b) distribution diagram of residual stress; (c) residual stress distribution along path I; (d) residual stress distribution along path II; (e) brazed joint after thinning the M42 steel.

Generally, ceramics have a low CTE, while M42 steel and the brazing seam have significantly higher values [5]. Thus, compressive stress is expected near the ceramic interface after brazing. However, the measured residual stress distribution showed tensile stress was presented at the ceramic interface. Despite of the CTE mismatch, residual stress in dissimilar joint could also arise from phase transformation and non-uniform heating. The brazing process was carried out in a vacuum furnace with moderate heating and cooling speed. Thus, there should be little contribution to the residual stress from the nonuniform heating. Considering that M42 high-speed steel contains a significant amount of carbon, it is speculated that this tensile stress arises from volume expansion caused by phase transformation in the steel. To verify this hypothesis, high-temperature XRD tests were performed on M42 steel and M42 steel with melted CuNi layer. Above 1000°C, M42 steel remains austenitic with carbides. Below 1000°C, austenite gradually transforms into ferrite, completing the transformation by 800°C. In contrast, the CuNi layer undergoes no phase transformation during cooling. Fig. 7(c) and (d) illustrate CTE curves of M42 steel and the separated CuNi layer. At a heating rate of 5°C/min, M42 steel undergoes a ferrite-to-austenite transformation around 800°C, causing a sudden volume shrinkage. Upon further heating, austenite expands, and the CTE gradually becomes positive. The maximum CTE of the steel reaches $30.3 \times 10^{-6}/^{\circ}\text{C}$ at 827°C. Beyond this point, the CTE decreases slowly, likely due to residual ferrite phase transformations and the dissolution of carbon [35]. Meanwhile, the CTE of the separated CuNi layer remains positive and increases steadily with temperature. XRD analysis of the original M42 steel, M42 steel with melted CuNi layer, and M42 steel after brazing revealed microstructural changes. Original M42 steel contained ferrite and carbides, while M42 steel with a melted CuNi layer showed ferrite, carbides, and retained austenite due to rapid furnace cooling. After brazing, M42 steel fully transformed into ferrite and carbides due to the slower

cooling rate of 5°C/min, allowing complete phase transformation [36]. Additionally, XRD patterns of the CuNi layer on M42 steel and the separated CuNi layer showed an increase in the 2 θ value after separation, indicating reduced interplanar spacing (d) and residual stress from brazing.



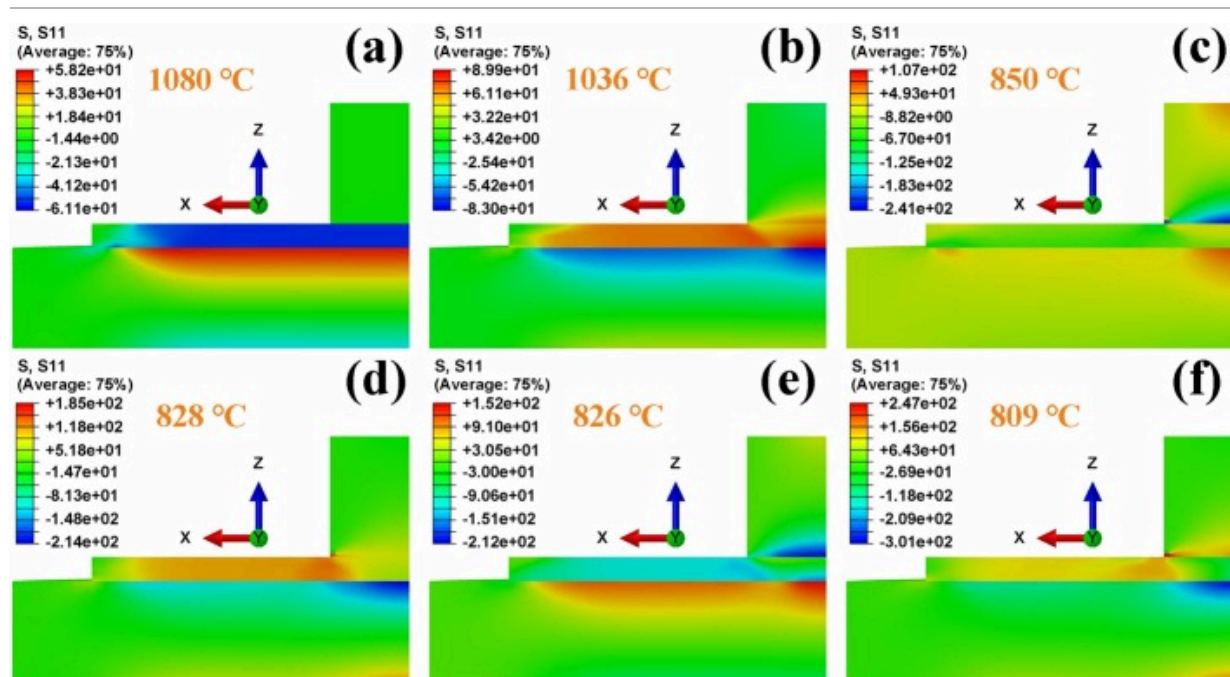
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Fig. 7. (a) XRD pattern of M42 steel at temperatures ranging from 1000 to 600°C; (b) XRD pattern of the CuNi layer at temperatures ranging from 1000 to 600°C; (c) CTE of M42 steel; (d) CTE of the separated CuNi layer; (e) XRD patterns of M42 steel in the original state, after melting CuNi layer, and after brazing; (f) XRD patterns of the CuNi layer melted onto the steel and the separated layer.

Therefore, the tensile stress at the ceramic interface is hypothesized to arise as follows: during brazing, M42 steel consists mainly of austenite and carbides. As cooling reaches the phase transformation point, austenite converts to ferrite, causing linear expansion [37]. The CuNi layer, with lower yield strength, extends with the steel. As cooling continues, the shrinkage of the CuNi interlayer and M42 steel is accommodated through slow plastic deformation. This imbalance leaves the ceramic side in tensile stress after cooling.

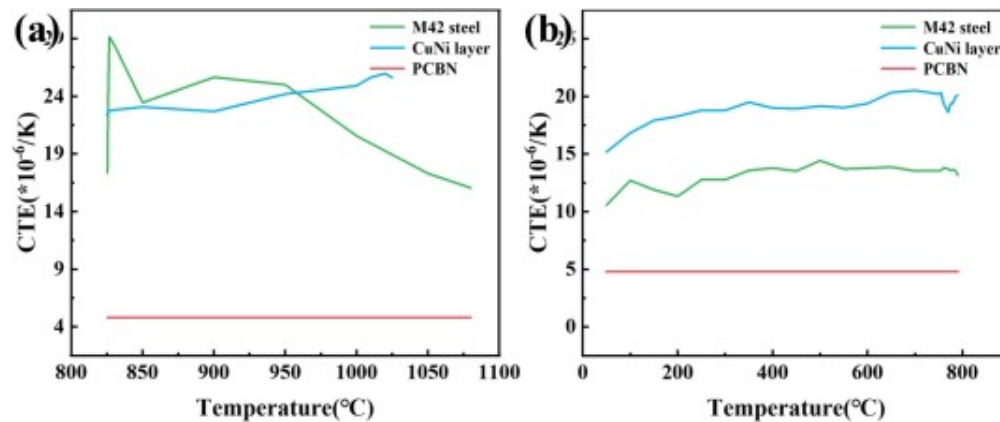
In the vacuum furnace, the joint experiences no horizontal constraints and cools slowly ($5^{\circ}\text{C}/\text{min}$), minimizing redistribution of residual stresses due to external constraints or pronounced temperature gradients. We simulated the joint's residual stress via the finite element method using ABAQUS software, with results presented in Fig. 8. The model size is consistent with the actual sample (Fig. S2), and material properties are listed in Table S2. At the isothermal solidification temperature, the PCBN ceramic, CuNi layer, and M42 steel exhibit a low-stress state (Fig. 8a). Residual stresses in the CuNi layer and M42 steel arise from uneven deformation during the pre-melting and bonding heating stages, with details illustrated in Figs. S3 and S4. As the temperature decreases prior to the steel's phase transition ($\sim 825^{\circ}\text{C}$), residual tensile stress does develop at the ceramic interface due to mismatches in thermal expansion coefficients (Fig. 9a), elastic moduli, and yield stresses. However, these stresses are minimal and do not significantly influence the final residual stress distribution. Within the phase transition temperature range (Fig. 8f), a significant tensile stress ($\sim 200\text{MPa}$) develops at the ceramic interface—though this value is substantially lower than the measured result. This discrepancy is likely attributable to the omission of deformation rate effects on the yield strength of the CuNi layer.



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Fig. 8. Finite element simulation of residual stress distribution in PCBN/CuNi/M42 joints during cooling above the M42 steel phase transition temperature: (a) 1080 °C; (b) 1036 °C; (c) 850 °C; (d) 828 °C; (e) 825 °C; (f) 809 °C.

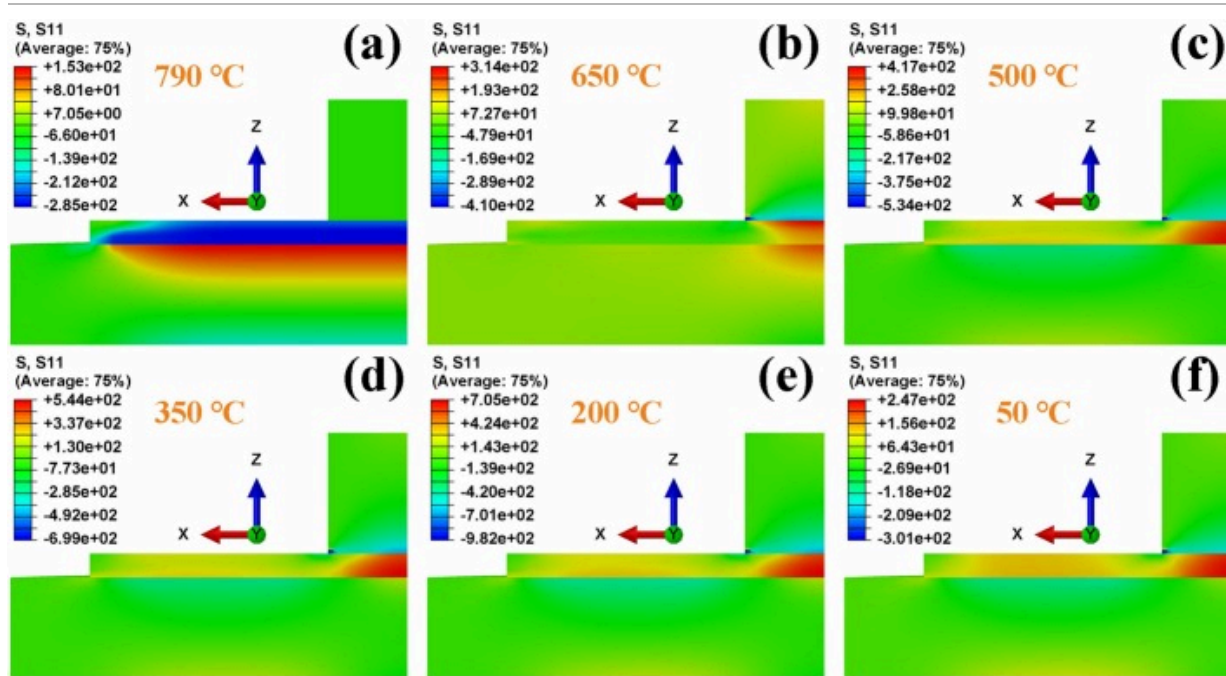


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Fig. 9. (a) Thermal expansion coefficients of PCBN, CuNi interlayer, and M42 steel in the range of 825–1080 $^{\circ}C$; (b) thermal expansion coefficients of PCBN, CuNi interlayer, and M42 steel in the range of 50–790 $^{\circ}C$.

In a simplified model assuming PTLP joint formation at 790 $^{\circ}C$, differences persist in the thermal expansion coefficients (Fig. 9b), Young's modulus, and yield strength among PCBN, the CuNi layer, and M42 steel. Yet residual stress is no longer influenced by phase transformation. Simulation results (Fig. 10) show that as the temperature falls below the phase transition threshold, dominant shrinkage of the CuNi layer leads to residual compressive stress at the ceramic interface.



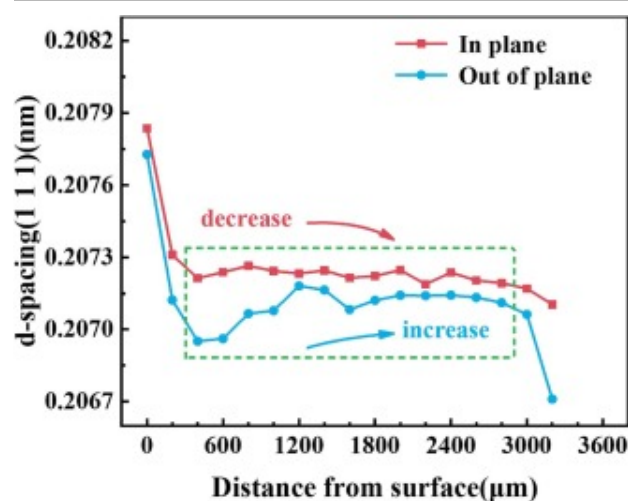
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Fig. 10. Finite element simulation of residual stress distribution in PCBN/CuNi/M42 joints during cooling under the M42 steel phase transition temperature: (a) 790 °C; (b) 650 °C; (c) 500 °C; (d) 350 °C; (e) 200 °C; (f) 50 °C.

The CuNi barrier layer exhibits a high thermal expansion coefficient and plastic deformability, combined with low yield strength at elevated temperatures. During sample cooling, this layer contracts continuously, theoretically exerting a compressive effect on the adjacent ceramic at the interface. However, comparison of in-plane and out-of-plane interplanar spacings along Path I (Fig. 11) reveals that the in-plane spacing consistently exceeds the out-of-plane spacing, indicating that the in-plane stress must be tensile. Within the 400–2800 μm range from the interface (toward the surface), the in-plane spacing gradually decreases while the out-of-plane spacing increases from the surface to the interface. This trend suggests a gradual reduction in in-plane residual tensile stress,

consistent with the residual stress measurements along Path I. Creep effects on residual stress are considered negligible, given the low creep strength and rate of the CuNi alloy at high temperatures. With no additional in-plane constraints in the lateral direction, the observed residual tensile stress is attributed to phase transformation of the M42 steel. Reference has indicated that copper exhibits higher high-temperature yield strength under high strain rates due to dislocation-phonon interactions [38]. Given that M42 steel undergoes a larger volume strain rate during phase transformation, the influence of the CuNi barrier layer on residual stress near the PCBN interface unfolds as follows: During phase transformation of M42 steel, its volume expansion is transmitted to the PCBN ceramic through the CuNi layer at an elevated strain rate, inducing significant in-plane residual tensile stress in the ceramic. Subsequently, during slow cooling, contraction of the CuNi layer alleviates a portion of this tensile stress, resulting in reduced residual tensile stress at 400 μm from the interface. However, within 200 μm of the interface, significant in-plane residual tensile stress persists due to the formation of complex intermetallic compounds and the Poisson effect in the vicinity of the interface.



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Fig. 11. Changes in the in-plane and out-of-plane d-spacing of the (111) crystal plane from the surface to the interface.

Another critical consideration is the ability of PCBN ceramics to withstand significant residual tensile stresses. As no prior reports exist on the tensile strength of polycrystalline cubic boron nitride (PCBN), we have compiled relevant data to validate the rationality of the residual stress values observed in this study. The theoretical tensile and shear strengths of cubic boron nitride (cBN) are 66GPa and 39GPa [39,40], respectively. However, accounting for inherent crystal defects in PCBN ceramics, its actual tensile strength is typically on the order of GPa. In addition, the tensile strengths of Al_2O_3 and SiC ceramics are reported to reach 300MPa and 500MPa [41,42], respectively. In this study, the region with residual tensile stress exceeding 1000MPa is localized within 200 μm of the interface or 400 μm of the surface, representing a localized stress concentration rather than bulk stress. Such local stresses are unlikely to exceed the local tensile strength of PCBN. Supporting this, previous studies have reported residual tensile stresses of $\sim 800\text{MPa}$ in sapphire joints without compromising joint integrity [43]. Additionally, finite element simulations of PCBN brazed joints have predicted local tensile stresses up to 2000MPa [44], further indicating that such levels are mechanically tolerable. Given the high hardness and Young's modulus of PCBN ceramics, their local tensile strength is sufficient to withstand residual tensile stresses of $\sim 1000\text{MPa}$, consistent with the observations in this work.

Moreover, a direct correlation exists between the measured residual stress values and the joint shear strength. Comparative analysis of PTLP joints with varying CuNi barrier layer thicknesses reveals that higher residual tensile stress near the interface corresponding to lower shear strength, indicating that increased interfacial tensile stress detrimentally affects the joint's load-bearing capacity. Thus, effective management of this stress is crucial for reliable high-temperature brazing between PCBN ceramics and M42 steel.

4. Conclusion

In conclusion, high-strength joining of PCBN ceramics and M42 steel was achieved using PTLP method. The Cu—Ni barrier layer minimized Ti diffusion and unwanted compounds, forming a near-solid solution joint. Synchrotron XRD measured the 2D residual stress distribution, revealing symmetric stresses—tensile at the surface/interface and compressive near the central axis and edges. The maximum tensile stress ($>1000\text{MPa}$) at the interface impacts joint strength and fracture. High-temperature XRD and CTE tests confirm that M42 steel's phase transformation and volume expansion cause this stress. These findings aid in controlling residual stress and improving ceramic-metal brazed joints.

CRediT authorship contribution statement

Lei Chen: Writing – original draft, Validation, Software, Investigation, Formal analysis, Data curation. **Chun Li:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation. **Mushi Zheng:** Methodology, Investigation, Data curation. **Bo Sun:** Visualization, Supervision, Software. **Ruhao Zhou:** Software, Methodology, Investigation. **Xiaoqing Si:** Visualization, Software, Methodology. **Bo Yang:** Software, Methodology, Data curation. **Haoran Yang:** Validation, Investigation, Data curation. **Jian Cao:** Writing – review & editing, Supervision, Resources, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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Appendix A. Supplementary data

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Supplementary material

Data availability

Data will be made available on request.

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